

Calculate Mean, Median, Mode and Range of the following data.

a 3, 6, 7, 8, 8

b 2, 5, 5, 7, 8

c -1, -1, -2, -4, -7

d 0.3, 0.6, 0.7, 0.8, 0.8, 0.9

e -1, -1, 2, 4, 5, 7

f 2.7, 2.9, 2.5, 2.8, 2.7, 2.5, 2.2, 2.6



Scenario: Tracking **Daily Temperatures** over one week.

The Data: 28°C, 29°C, 28°C, 30°C, 28°C, 31°C, 29°C.

The Question: What is the average temperature for the week?

- Since the temperatures are all relatively close together, the **Mean** provides a very precise and useful summary.

Real-Life Usage: Use the **Mean** when data is consistent and distributed evenly without massive jumps.



Scenario: A shoe store owner looking at **Shoe Sizes Sold**.

The Data: Sizes 7, 8, 8, 8, 9, 10, 12.

The Question: Which size should the owner order the most of for next month?

- The **Mean** might be 8.8, but you can't buy a size 8.8 shoe. The **Mode** (Size 8) tells the owner exactly what the customers want.

Real-Life Usage: Businesses use the **Mode** to manage inventory and stock the items that sell most frequently.



Organizing a class party based on **Favorite Types of Food**.

The Data: Pizza, Pasta, Pizza, Sushi, Pizza, Burgers, Pizza, Salad.

The Question: How do we decide what to order?

- You can't add "Pizza" and "Sushi" to find an average. You must use the **Mode**.

Real-Life Usage: Use the **Mode** for **Categorical Data** (words instead of numbers) to find the most popular choice.



Scenario: Analyzing **Daily Screen Time** (Mobile, TV, or Gaming).

The Data: 1 hr, 1.5 hrs, 2 hrs, 1.5 hrs, 8 hrs (a rainy Saturday!), 1 hr, 2 hrs.

The Question: Does the **Mean** or **Median** better represent a "normal" day?

- The **Mean** is pulled higher by the 8-hour outlier. The **Median** (1.5 hrs) is a more accurate "typical" day.
- **Real-Life Usage:** Use the **Median** when you have extreme values (outliers) that don't represent the norm.

